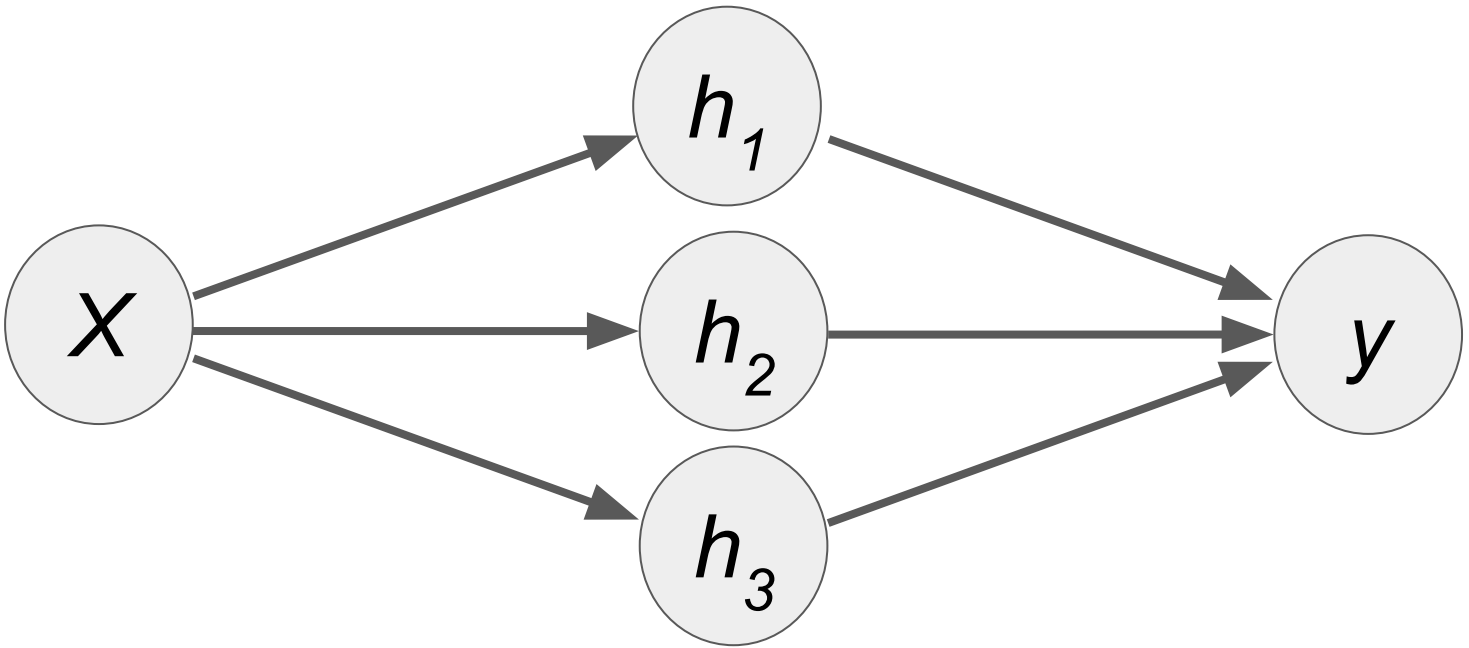
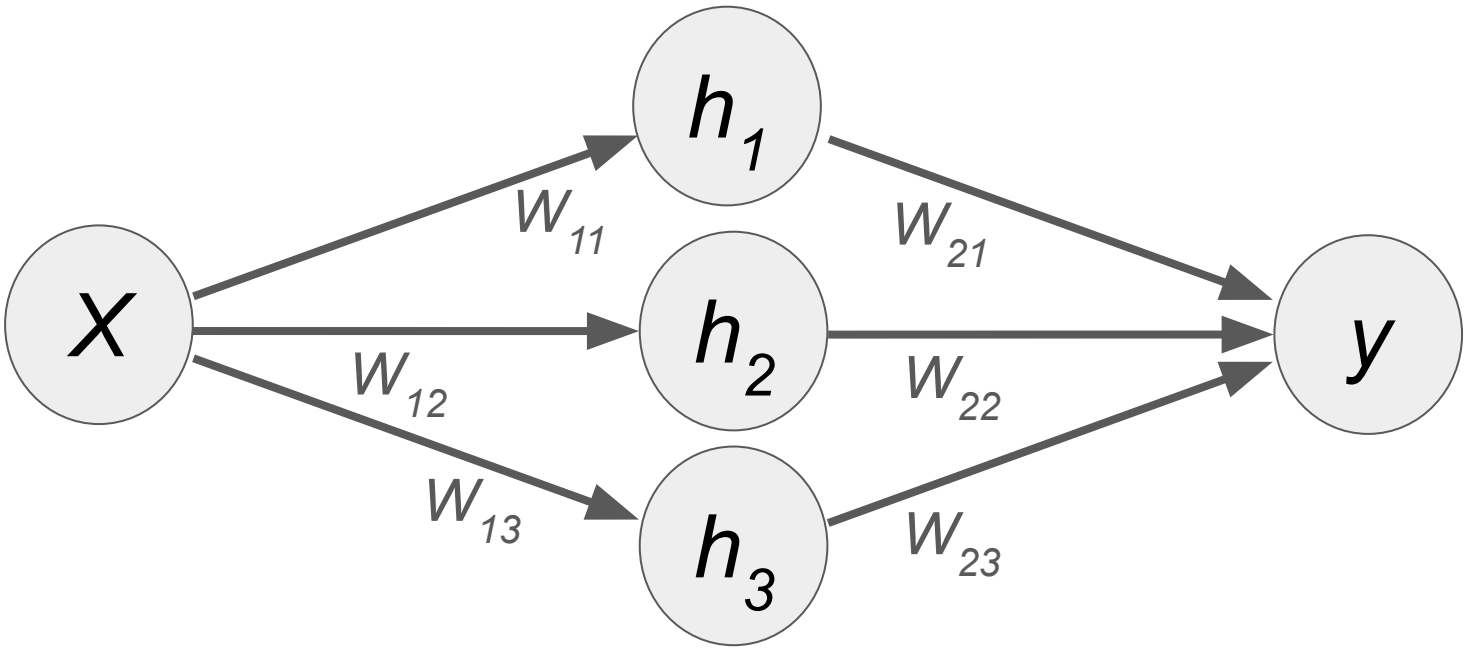
Input Hidden Layer Output

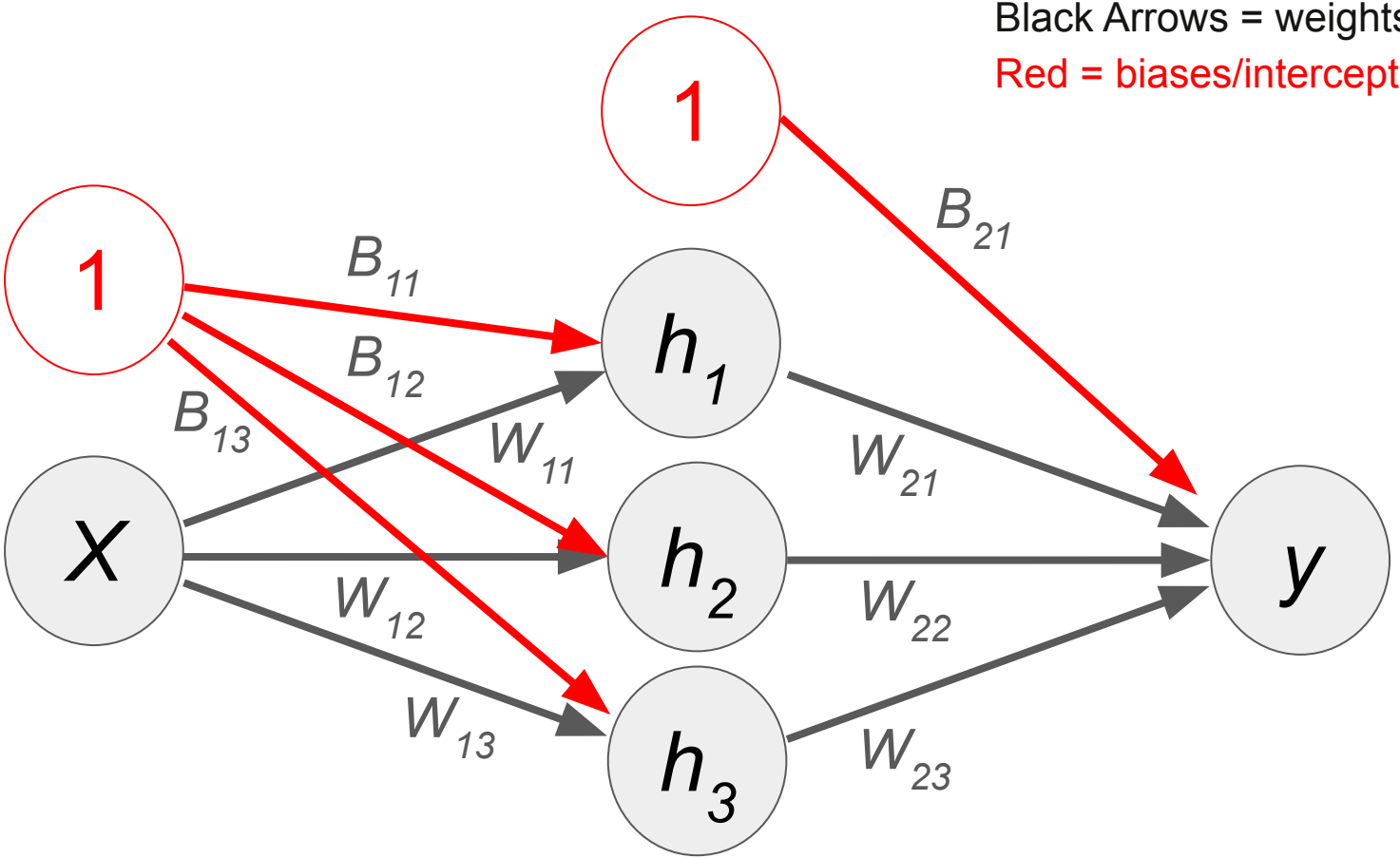


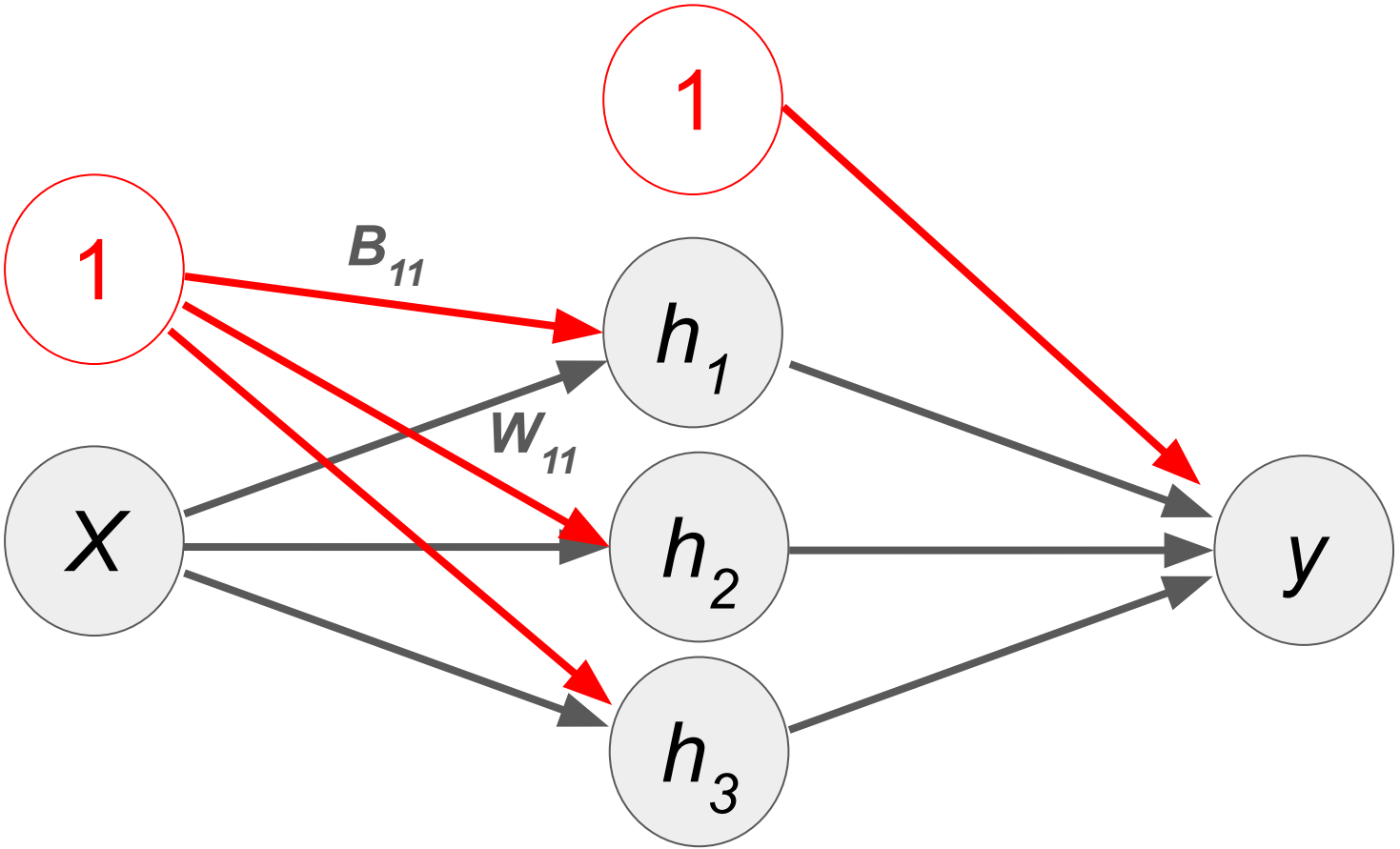
Each arrow represents a parameter

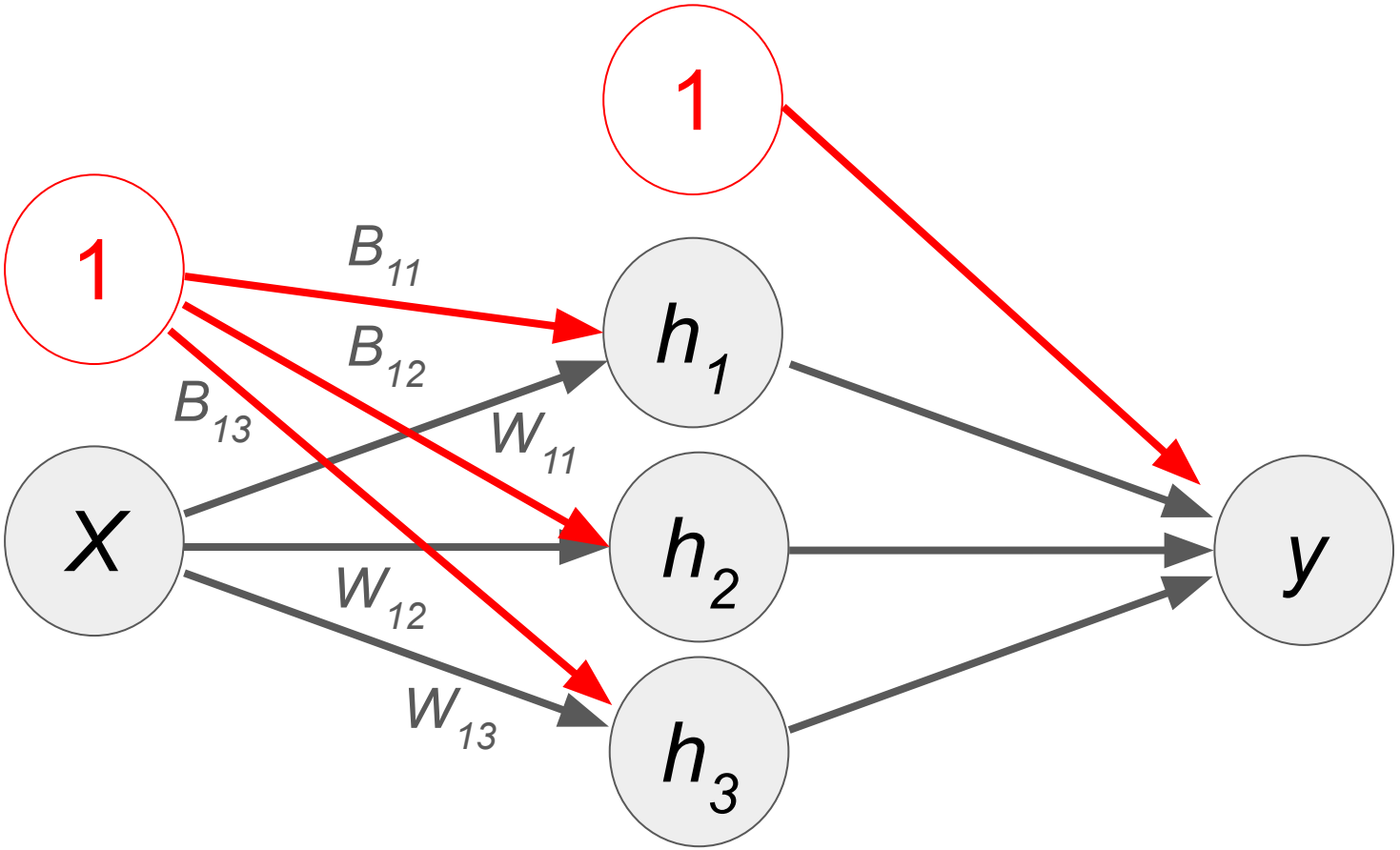


Black Arrows = weights/slopes

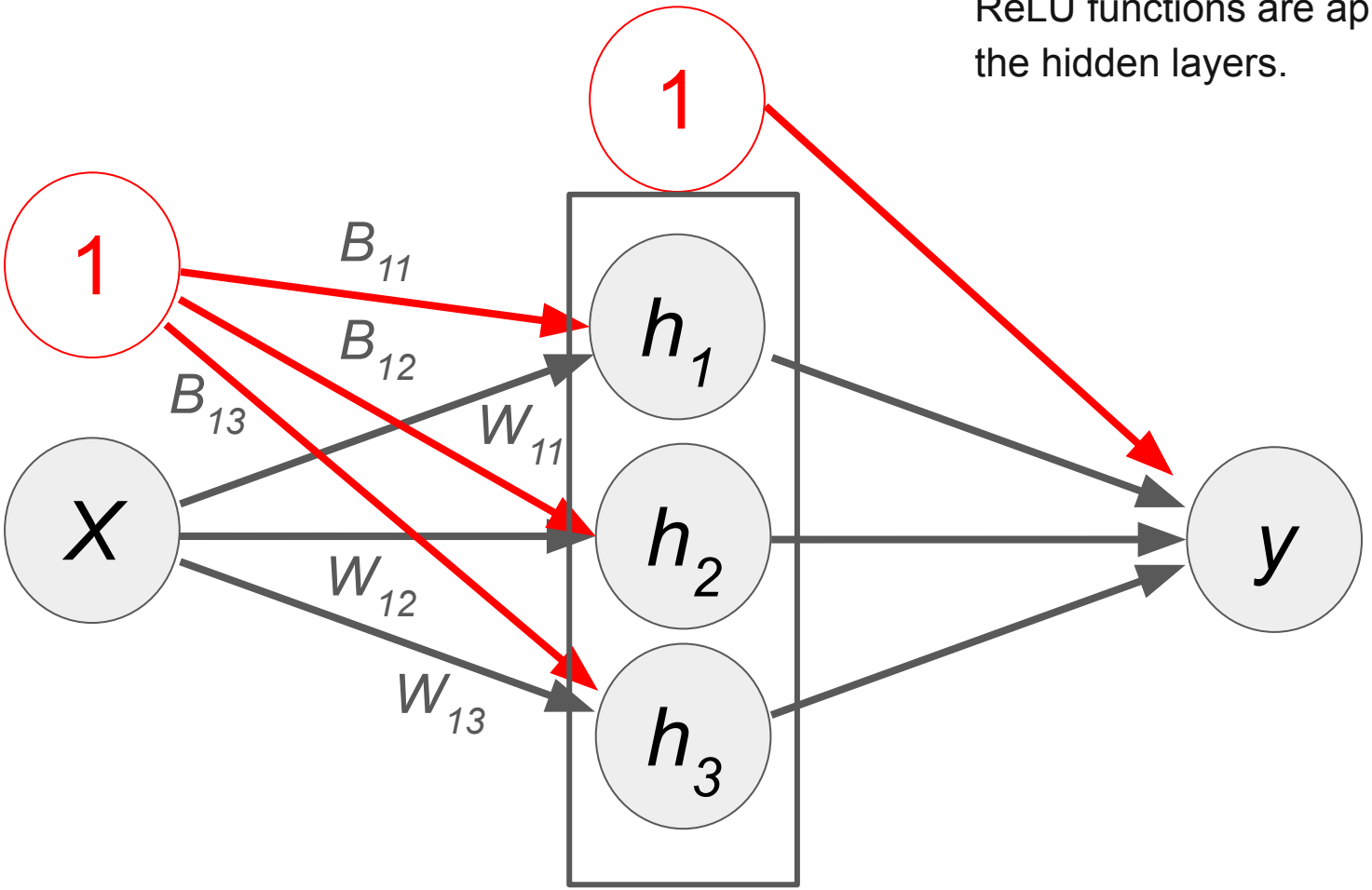
Red = biases/intercepts

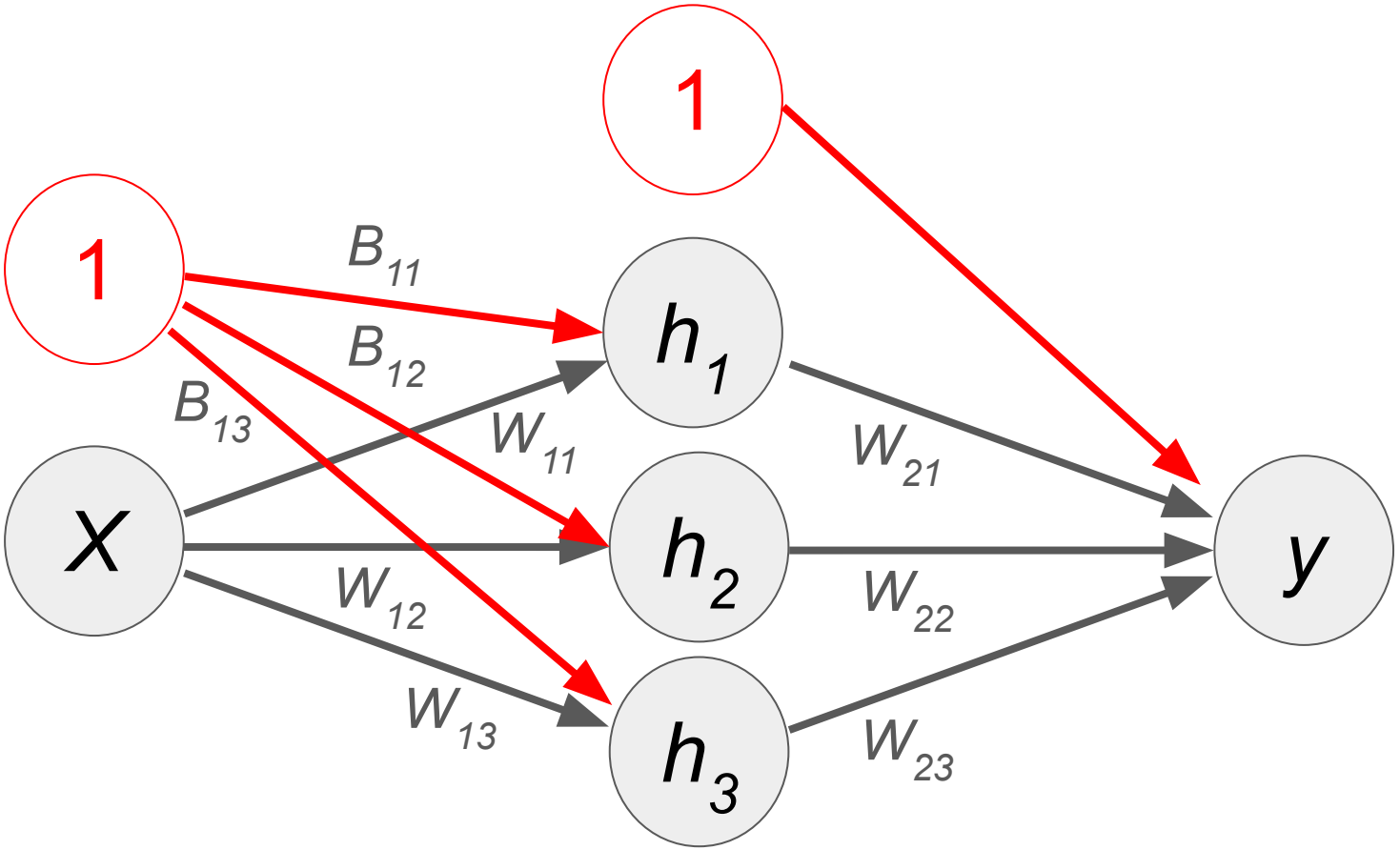


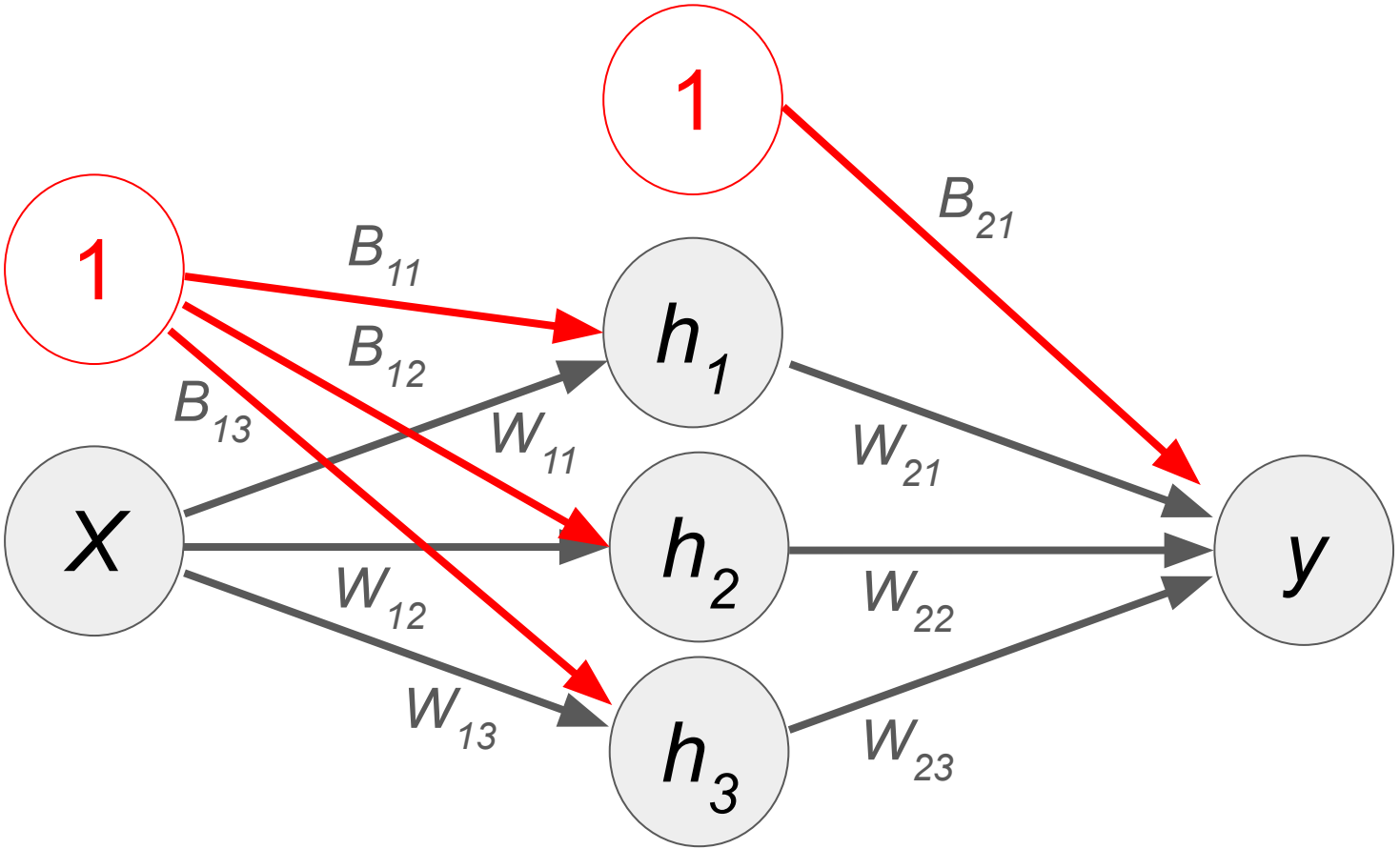

1
1
B_{11}
W_{11}
X
h_1
h_2
h_3
y

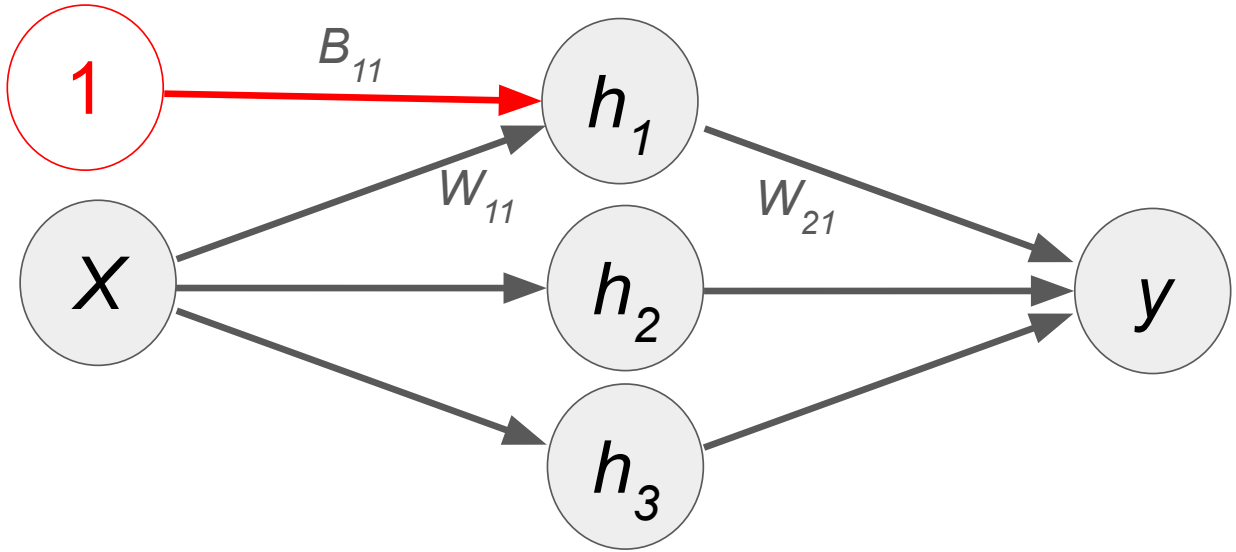

1
1
B_{11}
B_{12}
B_{13}
W_{11}
X
W_{12}
W_{13}
h_1
h_2
h_3
y

ReLU functions are applied at the hidden layers.




1
1
B11
B12
B13
W11
W12
W13
X
h1
h2
h3
W21
W22
W23
y


1
1
X
B_{11}
B_{12}
B_{13}
B_{21}
W_{11}
W_{12}
W_{13}
W_{21}
W_{22}
W_{23}
h_1
h_2
h_3
y



Forward Pass:

1. $Z_1 = B_{11} + W_{11} * X$
2. $h_1 = ReLU(Z_1)$
3. $y_hat = W_{21} * h_1$

Loss Calculation:

4. $L = (y - y_hat)^2$

Derivatives for backprop:

1. **$(dZ_1 / dW_{11}) = X$; $(dZ_1 / dB_{11}) = 1$;**
2. $(dh_1 / dZ_1) = 1$ *if* $Z_1 > 0$, *else* 0
3. **$(dy_hat / dW_{21}) = h_1$**; $(dy_hat / dh_1) = W_{21}$;

Loss Calculation:

4. $(dL / dy_hat) = -2 * (y - y_hat)$